Rebuttal for 3D Reconstruction of Interacting Multi-Person in Clothing from a Single Image

We thank the reviewers for their insightful comments. We will address all the editorial suggestions including typos; misleading statements, terms, and notations; and more analysis. We will cite all the missing related works. We will release our code.

[R1, R3] Deeper analysis on the impact of the inaccurate SMPL estimation and refinement module; comparison against DMC. We perform an ablation study in Fig. 1 where the reconstruction error is highly suppressed by optimizing the 3D pose with surface contact and inter-penetration priors. Based on the comparison with DMC in Tab. 1, we highlight that our refinement module is compatible with any off-the-shelf mesh reconstruction method to improve its accuracy. However, it demonstrates weak performance when there exists a strong occlusion as shown in Fig. 2, which leads to the failure of contact estimation and inaccurate 3D mesh reconstruction.

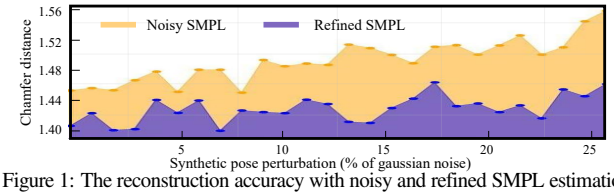

Figure 1: The reconstruction accuracy with noisy and refined SMPL estimation. The noisy pose is synthetically generated by adding the fraction of Gaussian noise which perturbs ground truth SMPL poses.

	Two natural-inter		Two close-inter		Three people	
Method	init.	ref.	init.	ref.	init.	ref.
DMC	2.53	2.50	3.24	3.10	3.81	3.76
Ours	1.19	**1.17**	1.84	**1.34**	1.76	**1.54**

Table 1: Comparison with DMC on MultiHuman dataset based on Chamfer Distance metric. 'init.' and 'ref.' denote initial SMPL from [4] and refined SMPL.

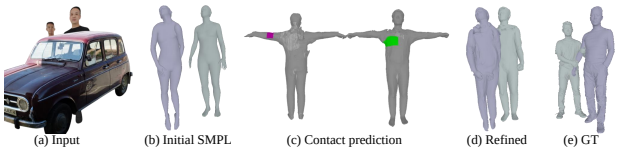

Figure 2: Failure case: a strong occlusion leads to inaccurate contact prediction (left arm and right chest) and inaccurate 3D reconstruction results.

[R1] Which SMPL estimation method used in Fig. 5 and Fig. 6. We used the method from Cha et al. [4] for DMC in Fig. 6 and ours in Fig. 5 (b). We used the refined SMPL for ours in Fig. 6 and Fig. 5 (c-e) optimized with corresponding losses.

[R2, R3] How does the missing "z_{detail}" affect the level of details? Original gDNA [5] used z_{detail} to generate random residual texture in 2D space to augment geometric details. Instead of predicting random 2D texture, we separately design an implicit function f^{agg} that can combine pixel-aligned features from the observed surface normal, which allows generating physically correct and coherent 3D surface residuals.

[R2] What is the performance gained by adding $\mathbf{F}^{\text{2D}}$? In Tab. 3 (Ours w/o N) of the main paper, adding $\mathbf{F}^{\text{2D}}$ improves the 3D reconstruction quality as much as 0.16 of Chamfer distance on average and 0.27 of Point-to-Surface distance on average.

[R2] What is the baseline performance of just predicting z_{shape}? In Tab. 2, we summarize the performance of the suggested baseline method where our full model consistently outperforms the baseline by generating accurate geometric details and globally coherent mesh with the average gain of 0.21 Chamfer distance.

Method	Single	Occluded single	Two natural-inter	Two closely-inter	Three	Avg.
Only z_{shape}	1.41	1.66	1.28	1.91	1.82	1.61
Ours	**1.37**	**1.63**	**1.17**	**1.33**	**1.54**	**1.40**

Table 2: Comparison of our method with the one that just predicts z_{shape} on MultiHuman dataset based on Chamfer distance metric.

[R2] The results look mostly as nearest-neighbours of gDNA with missing subject-specific details. We acknowledge that the geometric diversity is limited due to the fundamental domain constraints of 3D human geometric prior, *i.e.*, gDNA, as noted in the conclusion. However, we kindly point out that high-frequency details are beyond gDNA where our implicit network generates subject-specific geometric details by combining the pixel-aligned features as shown in Fig. 4 of the main paper.

[R2, R3] What exactly makes it "complete" robust to people-people occlusion and "globally coherent"? The active use of human geometry prior (gDNA) makes the results "complete": our new encoder predicts the latent space of a 3D "complete" human that best matches a 2D occluded person image. The effective 3D pose optimization based on our surface contact and inter-penetration priors enables the 3D reconstruction of people in the "globally coherent" world coordinates.

[R3] Relationship between the proposed method and gDNA [5]. Our new encoder regresses an occluded person image to predict the latent space (gDNA) of a complete human. Our new implicit network generates person-specific and physically plausible geometric details by combining the decoded gDNA features and the pixel-aligned features.

	[8]	f^{Contact} (ours)
Accuracy ↑	88%	**96%**

Table 3: Evaluation on our contact discriminator f^{Contact}.

[R3] Evaluation on the contact estimation networks is required. In Tab. 1 of the supplementary manuscript, f^{CS} outperformed [8] on $\text{IoU}_{\text{Segm.}}$ by 0.03 and $\text{IoU}_{\text{Signature}}$ by 0.09. In Tab. 3, we perform additional comparison to evaluate f^{Contact} where f^{Contact} outperforms [8] by 8% accuracy. We will add the above tests to the main paper with details and analysis.

[R3] The rationale for having 75 body regions. How is the contact signature defined? Each body region is grouped by following the human body anatomy and symmetries [8]. The contact signature is defined as 75×75 matrix for every two-people.

[R3] Having both f^{Contact} and f^{CS} is redundant. Both f^{Contact} and f^{CS} are needed because they are complementary. f^{Contact} regresses an input image to the binary value to classify if contacts exist in the scene, and f^{CS} predicts the surface relation over people only when the contact exists. Such complementary nature improves the 3D reconstruction accuracy as shown in Fig. 3.

Figure 3: The 3D reconstruction results without and with f^{Contact}.